Aptamers from random sequence space: Accomplishments, gaps and future considerations.

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Aptamers are molecular recognition elements made of nucleic acids. Diverse synthetic aptamers have been discovered in a large number of SELEX experiments since 1990. This review begins with the analysis of these SELEX experiments by examining the range of targets as well as the affinity and specificity for these targets by DNA, RNA or modified nucleic acid aptamers generated from these experiments. This is followed by comparisons of synthetic aptamers with natural RNA aptamers from riboswitches and with some of the best naturally occurring protein based recognition elements for proteins and small molecules. These comparisons reveal the gaps between man-made aptamers and natural recognition elements. We then put forward a series of ideas for consideration by the aptamer community towards developing better aptamers to solve real world problems. These include performing aptamer selections with libraries containing larger random sequence domains to derive larger aptamers with more intricate structures, conducting more reselection experiments using mutagenized DNA libraries based on initially selected aptamer sequences to search for better members of an aptamer family, selecting aptamers that are programmed to recognize two or more different epitopes of the same target in order to build multivalent aptamers for increased affinity, expanding the effort of selecting aptamers using modified nucleic acids with enhanced chemical functionalities, innovating SELEX methods to drive for the selection of aptamers with truly outstanding affinity and specificity, and having terminal applications in mind when creating new aptamers so that they are highly functional in the environment where the applications are planned.

The chemical biology of aptamers.

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Aptamers are small single-stranded nucleic acids that fold into a well-defined three-dimensional structure. They show a high affinity and specificity for their target molecules and inhibit their biological functions. Aptamers belong to the nucleic acids family and can be synthesized by chemical or enzymatic procedures, or a combination of the two. They can, therefore, be considered as both chemical and biological substances. This Review summarizes the most convenient approaches to their preparation and new developments in the field of aptamers. The application of aptamers in chemical biology is also discussed.

The future of aptamers in medicine.

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Aptamers, simply described as chemical antibodies, are synthetic oligonucleotide ligands or peptides that can be isolated in vitro against diverse targets including toxins, bacterial and viral proteins, virus-infected cells, cancer cells and whole pathogenic microorganisms. Aptamers assume a defined three-dimensional structure and generally bind functional sites on their respective targets. They possess the molecular recognition properties of monoclonal antibodies in terms of their high affinity and specificity. The applications of aptamers range from diagnostics and biosensing, target validation, targeted drug delivery, therapeutics, templates for rational drug design to biochemical screening of small molecule leads compounds. This review describes recent progress made in the application of biomedically relevant aptamers and relates them to their future clinical prospects.

Aptamers as therapeutic middle molecules.

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Therapeutic molecules can be classified as low-, middle- and high-molecular weight drugs depending on their molecular masses. Antibodies represent high-molecular weight drugs and their clinical applications have been developing rapidly. Aptamers, on the other hand, are middle-molecular weight molecules that are short, single-stranded nucleic acid sequences that are selected in vitro from large oligonucleotide libraries based on their high affinity to a target molecule. Hence, aptamers can be thought of as a nucleic acid analog to antibodies. However, several viewpoints hold that the potential of aptamers arises from interesting characteristics that are distinct from, or in some cases, superior to those of antibodies. Recently, therapeutic middle molecules gain considerable attention as protein-protein interaction (PPI) inhibitors. This review summarizes the recent achievements in aptamer development in our laboratory in terms of PPI and non-PPI inhibitors.

Aptamers: current challenges and future prospects.

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Aptamers are oligonucleotide molecules raised in vitro from large combinatorial libraries of nucleic acids and developed to bind to targets with high affinity and specificity. Whereas novel target molecules are proposed for therapeutic intervention and diagnostic, aptamer technology has a great potential to become a source of lead compounds. In this review, the authors address the current status of the technology and highlight the recent progress in aptamer-based technologies. They also discuss the current major technical limitations of aptamer technology and propose original solutions based on existing technologies that could result in a solid aptamer-discovery platform. Whereas aptamers have shown to bind to targets with similar affinities and specificities to those of antibodies, aptamers have several advantages that could outweigh antibody technology and open new opportunities for better medical and diagnostic solutions. However, the current status of the aptamer technology suffers from several technical limitations that slowdown the progression of novel aptamers into the clinic and makes the business around aptamers challenging.

Current Perspectives on Aptamers as Diagnostic Tools and Therapeutic Agents.

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Aptamers are synthetic single-stranded DNA or RNA sequences selected from combinatorial oligonucleotide libraries through the well-known in vitro selection and iteration process, SELEX. The last three decades have witnessed a sudden boom in aptamer research, owing to their unique characteristics, like high specificity and binding affinity, low immunogenicity and toxicity, and ease in synthesis with negligible batch-to-batch variation. Aptamers can specifically bind to the targets ranging from small molecules to complex structures, making them suitable for a myriad of diagnostic and therapeutic applications. In analytical scenarios, aptamers are used as molecular probes instead of antibodies. They have the potential in the detection of biomarkers, microorganisms, viral agents, environmental pollutants, or pathogens. For therapeutic purposes, aptamers can be further engineered with chemical stabilization and modification techniques, thus expanding their serum half-life and shelf life. A vast number of antagonistic aptamers or aptamer-based conjugates have been discovered so far through the in vitro selection procedure. However, the aptamers face several challenges for its successful clinical translation, and only particular aptamers have reached the marketplace so far. Aptamer research is still in a growing stage, and a deeper understanding of nucleic acid chemistry, target interaction, tissue distribution, and pharmacokinetics is required. In this review, we discussed aptamers in the current diagnostics and theranostics applications, while addressing the challenges associated with them. The report also sheds light on the implementation of aptamer conjugates for diagnostic purposes and, finally, the therapeutic aptamers under clinical investigation, challenges therein, and their future directions.

[Aptamers in clinical diagnostics].

PMID: 17201061 | Indexed in PubMed

Aptamers are single-stranded DNA or RNA oligonucleotides selected in vitro from combinatorial libraries in a process called SELEX (Systematic Evolution of Ligands by EXponential Enrichment). Aptamers play a role of artificial nucleic acid ligands that can recognize and bind to various organic or inorganic target molecules with high specificity and affinity. They can discriminate even between closely related targets and can be easily chemically modified for radioactive, fluorescent and enzymatic labeling or biostability improvement. Aptamers can thus be considered as universal receptors that rival antibodies in diagnostics as a tool of molecular recognition. To date aptamers have been successively used instead of monoclonal antibodies in flow cytometry, immunochemical sandwich assays and in vivo imaging as well to detect wide range of small or large biomolecules.

Aptamers in the Therapeutics and Diagnostics Pipelines.

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Aptamers are short single-stranded DNA or RNA oligonucleotides that can selectively bind to small molecular ligands or protein targets with high affinity and specificity, by acquiring unique three-dimensional structures. Aptamers have the advantage of being highly specific, relatively small in size, non-immunogenic and can be easily stabilized by chemical modifications, thus allowing expansion of their diagnostic and therapeutic potential. Since the invention of aptamers in the early 1990s, great efforts have been made to make them clinically relevant for diseases like macular degeneration, cancer, thrombosis and inflammatory diseases. Furthermore, owing to the aforementioned advantages and unique adaptability of aptamers to point-of-care platforms, aptamer technology has created a stable niche in the field of in vitro diagnostics by enhancing the speed and accuracy of diagnoses. The aim of this review is to give an overview on aptamers, highlight the inherent therapeutic and diagnostic opportunities and challenges associated with them and present various aptamers that have reached therapeutic clinical trials, diagnostic markets or that have immediate translational potential for therapeutics and diagnostics applications.

Aptamers and Their Significant Role in Cancer Therapy and Diagnosis.

PMID: 28536411 | Indexed in PubMed

Aptamers are nucleic acid/peptide molecules that can be generated by a sophisticated, well-established technique known as Systematic Evolution of Ligands by EXponential enrichment (SELEX). Aptamers can interact with their targets through structural recognition, as in antibodies, though with higher specificity. With this added advantage, they can be made useful for clinical applications such as targeted therapy and diagnosis. In this review, we have discussed the steps involved in SELEX process and modifications executed to attain high affinity nucleic acid aptamers. Moreover, our review also highlights the therapeutic applications of aptamer functionalized nanoparticles and nucleic acids as chemo-therapeutic agents. In addition, we have described the development of "aptasensor" in clinical diagnostic application for detecting cancer cells and the use of aptamers in different routine imaging techniques, such as Positron Emission Tomography/Computed Tomography, Ultrasound, and Magnetic Resonance Imaging.

Advances in aptamers.

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Aptamers are nucleic acid sequences synthesized through in vitro selection and amplification technique, possessing a broader range of applications in therapeutics, biosensing, diagnostics, and research. Aptamers offer a number of advantages over their antibodies counterpart, one of them is their ability to undergo chemical derivatization to increase their life in the body fluids and bioavailability in animals. Although aptamers were discovered in 1990s, they have become one of the most widely investigated molecules, with a huge number of publications in the last decade. This article presents an overview of the advancements that have been made in aptamers. We mainly focused on articles published since 2005.